A3.1 Diversity of organisms

NAME	DATE	TOTAL
		_____ / 40

About 40 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

8 MARKS

1 **STATE** what each part of a binomial name shows. [A3.1.3] [2]

2 **STATE** the biological species concept. [A3.1.4] [1]

3 **STATE** the diploid chromosome number of humans and of chimpanzees. [A3.1.6] [1]

4 **DEFINE** genome. [A3.1.8] [1]

5 **LIST** three features used to arrange chromosomes in a karyogram. [A3.1.7] [3]

Section B: Understanding

13 MARKS

6 **EXPLAIN** why a horse and a zebra are classed as separate species, even though they can produce offspring. [A3.1.4] [2]

7 **OUTLINE** two difficulties in applying the biological species concept. [A3.1.4] [2]

8 **EXPLAIN** why it can be difficult to tell populations from species during speciation. [A3.1.5] [3]

9 **COMPARE AND CONTRAST** genome variation within a species and between species. [A3.1.8, A3.1.9] [3]

10 **OUTLINE** how sequencing is used now, and how it may be used in future. [A3.1.11] [3]

Section C: Working with data

13 MARKS

The values in question 11 are approximate published figures, for practice.

11 [A3.1.10]

ORGANISM	GENOME SIZE / MILLION BASE PAIRS
E. COLI	5
FRUIT FLY	140
HUMAN	3100
LUNGFISH	130000

a **CALCULATE** how many times larger the human genome is than the fruit fly genome. [1]

b **DISCUSS** whether genome size is linked to the complexity of an organism. [3]

c **SUGGEST** why a large genome may not contain many more genes. [1]

12 [A3.1.7]

A student studies a karyogram from a human cell. Chromosomes are in 22 numbered pairs, plus one X and one Y chromosome. There are three copies of chromosome 21.

a **DEDUCE** the sex of the individual, giving a reason. [1]

b **CALCULATE** the total number of chromosomes in this cell. [1]

c **EVALUATE** the use of length alone to identify chromosomes. [2]

13 [A3.1.7]

Human chromosome 2 has telomere sequences near its middle, a second inactive centromere, and a banding pattern matching two chimpanzee chromosomes.

a EXPLAIN how this supports the hypothesis of a chromosome fusion. [3]

b EXPLAIN why it is incorrect to say that humans evolved from chimpanzees. [1]

Section D: Longer answer

6 MARKS

14 DISCUSS the different ways of defining a species, and why no single definition works for every organism. [A3.1.2, A3.1.4, A3.1.5] [6]